

Proxying the 1987 Market Break Using Only S&P 500 Futures Prices

Part A. Research Report

How the 1987 market break is described in primary accounts

Authoritative contemporaneous and near-contemporaneous accounts treat the October 1987 “market break” as more than a large price decline: it is framed as a period in which market infrastructures, intermarket linkages, and liquidity provision mechanisms were stressed to (or past) their functional limits.

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The most influential post-event government study—the report of the Presidential Task Force on Market Mechanisms² chaired by Nicholas F. Brady³—explicitly argues that what had been treated as “separate” venues (stocks, stock index futures, and options) behaved economically as *one market* in normal times, but in mid-October 1987 that “one market” failed to function as a single integrated system. It attributes the severity of the episode to the inability of the segmented infrastructures and rules to absorb massive, one-sided selling pressure, and describes “intermarket” mechanisms—especially those that transmit buying and selling between segments—as becoming ineffective at critical times.⁴

The Brady Task Force also describes the episode in explicitly *microstructural* terms: the “market break” produced operational breakdowns and trading interruptions that functioned like improvised circuit breakers—e.g., clogged routing/execution systems, ad hoc halts, and communications congestion—reflecting “natural limits” to market liquidity and the system’s capacity to process one-sided flow.⁵

The U.S. Securities and Exchange Commission⁶ staff report (February 1988)—a major primary source written close to the event—emphasizes “extraordinary” volume and volatility but also stresses that the investigation is not simply about why prices fell; it is about reconstructing trading and diagnosing how market structures and derivative-related practices interacted with the crisis. Crucially, the report states that the market break “did not result in merely a dramatic one-time revaluation,” and points to persistent deterioration in market quality indicators (e.g., liquidity/continuity and spreads) after the event—language that is naturally read as “structural disruption” rather than a transient spike in volatility.⁷

The U.S. Commodity Futures Trading Commission⁸ final report similarly describes the episode as creating “severe strains” on the operational and financial control systems of securities and futures exchanges, while noting that outright system failures were avoided. It also cautions against over-simplified mechanical narratives (e.g., “cascade theory”) by suggesting they describe at most a short-term technical realignment rather than being a complete explanation of the market break.⁹

The U.S. Government Accountability Office¹⁰ report underscores intraday dysfunction and information uncertainty: on October 19–20 it reports widespread confusion, large order imbalances, delayed openings/halts, jammed communications, and the inability of derivatives market participants to determine prices in

underlying markets—conditions that translated into higher volatility but were fundamentally *market-functioning* problems. ¹¹

A later Federal Reserve staff history by Mark Carlson ¹² crystallizes the same distinction: the crash mattered not only for the magnitude of price declines, but because “market functioning was severely impaired,” and trading/clearing frictions interacted with the price decline to amplify stress. ¹³

Implication for your task: If “market break” is defined by these sources as a period of *market functioning impairment and linkage failure*, then a futures-only proxy should focus on identifying a **regime shift in price formation consistent with liquidity/processing constraints and discontinuous repricing**, rather than simply flagging the single largest realized volatility print. ¹⁴

Volatility vs structural disruption in market microstructure terms

The SEC staff report makes an unusually explicit conceptual point that aligns with your instruction not to equate “market break” with “highest volatility.” It notes that volatility is, in one sense, a “neutral” description—capturing how quickly prices react to information—yet it argues that volatility at extreme levels has adverse consequences through market-making risk, higher liquidity costs, and impaired intermediation capacity. ¹⁵

In microstructure language consistent with Brady/GAO/SEC framing, a “market break” is best viewed as a **structural disruption**: a change in the *mapping from trading pressure and information into prices* because liquidity provision, trading continuity, and intermarket transmission channels (arbitrage, hedging, and operational linkages) are impaired. The Brady report’s “one market” framing plus its emphasis on the failure of linkages and the emergence of ad hoc circuit breakers is essentially a description of a regime in which the normal price discovery mechanism is constrained by infrastructure and liquidity limits. ¹⁶

Quantitatively, “structural break” is commonly formalized as a change in the parameters governing a time series process (means, variances, autocorrelation structure, tail behavior, etc.), often at one or more changepoints. Multiple structural break frameworks in econometrics (e.g., regression parameter break models) and algorithmic changepoint detection methods treat **breaks as changes in the generating distribution or model parameters**, which is conceptually much closer to “regime shift” than to “peak volatility.” ¹⁷

Implication for your task: A defensible futures-only “market break” timestamp should correspond to an estimated **changepoint/regime-transition** in the intraday return process (or in robust features derived from it) reflecting a shift in dependence/discontinuity/continuity—not merely the maximum absolute return or volatility. ¹⁸

What price-only data can and cannot reveal about a “break”

You correctly note that historical work often emphasizes a breakdown in **futures-cash linkage**. The primary sources make this linkage central: Brady emphasizes that when cross-market mechanisms (e.g., those enabling index arbitrage and information transmission) became ineffective, markets could “disengage,” and trading constraints in one segment could propagate dysfunction elsewhere. ¹⁹ The SEC staff report likewise highlights the importance of coordination between stock and derivative index products

and discusses how futures price dynamics and discounts affected behavior in the cash market, especially around openings. ²⁰

But with only futures prices, you cannot directly observe: - the futures–cash basis and its persistence (the most direct linkage metric), or cash-market “stale index” effects; ²¹
- bid–ask spread widening, depth collapse, order book imbalance, or volume/participation withdrawal (classic microstructure stress metrics), because those require quotes, depth, or volume. ²²

What price-only data *can* reveal—often in ways consistent with “market functioning impairment”—falls into three categories:

Discontinuity / jump behavior. High-frequency econometrics provides tools to distinguish “continuous” diffusive variation from rare discontinuous jumps using only prices (e.g., realized variance vs bipower variation), and to detect jump arrival times (intraday) nonparametrically. ²³ Discontinuous repricing is a plausible footprint of stressed price discovery and liquidity scarcity—conditions explicitly described in 1987 primary accounts (halts, clogged systems, inability to determine prices). ²⁴

Serial dependence / autocorrelation regime shifts. Classic microstructure models show that transaction costs and bid–ask “bounce” can induce negative serial dependence in observed price changes; this insight supports using autocovariance/autocorrelation measures as price-only, indirect indicators of trading frictions. ²⁵ A regime shift in serial dependence (sign, magnitude, stability) is interpretable as a change in the relative importance of microstructure frictions vs information-driven/trend-driven price moves—exactly the kind of “structural” change your framing targets. ²⁶

Changepoints in the return distribution. Changepoint detection methods (offline and online) are designed to identify abrupt changes in the generative parameters of a series (mean, variance, distribution shape) and can be extended to multivariate feature streams. ²⁷ This aligns with the “regime shift” framing of a market break.

A crucial caution given your dataset constraints: price-only indicators can be contaminated by **data-quality artifacts** (missing minutes, stale last prices, recording gaps) that mechanically create apparent jumps or autocorrelation shifts. Because primary accounts highlight “stale” and delayed prices as part of the 1987 dysfunction (especially in cash markets), it’s essential to treat staleness both as (i) a possible genuine stress signature and (ii) a possible measurement artifact in your particular dataset. ²⁸

Candidate futures-only methods and critical evaluation

Below are the most defensible *price-only* candidates for your minute futures series, targeted to your constraints (no cash index, no volume, possible missing/stale blocks). Each is evaluated in the specific sense you requested: what it measures, why it could represent a break, defensibility for 1987, strengths/weaknesses, and timestamp precision.

Rolling autocorrelation breakdown (including AR(1) instability and Roll-type spread proxy)

What it measures. Rolling first-lag autocorrelation of returns (or price changes), rolling AR(1) coefficients, and related diagnostics (e.g., variance ratio–style serial dependence measures). A specific price-only liquidity proxy is the “Roll” intuition: negative first-order serial covariance can arise from bid–ask bounce and trading costs. ²⁵

Why it could represent a market break. If liquidity provision and trading frictions change abruptly—consistent with 1987 descriptions of constrained market-making and disrupted trading continuity—short-horizon serial dependence can shift in sign or magnitude (e.g., from mild negative bounce to extreme negative bounce, or to positive autocorrelation if one-sided order flow and slow adjustment dominate). ²⁹

Historically defensible for 1987. Primary accounts identify extreme stress, uncertainty, and segments becoming “disengaged,” along with trading/processing constraints that plausibly altered the microstructure of price formation. That is exactly the environment in which serial dependence patterns can become unstable. ³⁰

Strengths. Uses only prices; can be computed robustly on 1-minute data; easy to pair with changepoint methods to timestamp a regime transition. Grounded in canonical microstructure logic about how trading frictions leave signatures in serial covariance. ³¹

Weaknesses. Autocorrelation is not uniquely diagnostic of “dysfunction”: trending markets, discrete jumps, and time aggregation can generate autocorrelation changes unrelated to market breakdown. The Roll logic is especially fragile in strongly trending or jump-dominated intervals because the first-order covariance can turn positive (making the “spread” undefined) and because the assumptions behind the model are restrictive. ²⁵

Can it identify one exact break timestamp? Not by itself. Autocorrelation is usually estimated in windows; it is better used as an *input feature* to a changepoint detector, which can return a specific timestamp (typically at the boundary between segments, at your 1-minute resolution). ³²

Jump behavior and discontinuity measures (bipower variation decomposition; intraday jump timestamps)

What it measures. The proportion of realized variation attributable to jumps (e.g., realized variance minus bipower variation, scaled appropriately), and/or explicit detection of jump arrival times from high-frequency returns using nonparametric tests. ²³

Why it could represent a market break. A shift from “mostly continuous” price evolution to a process dominated by discrete jumps and rapid gaps is consistent with an environment of impaired liquidity and disrupted price discovery—conditions described in 1987 accounts (halts, inability to establish transactable prices, abrupt dislocations, and ad hoc trading interruptions). ²⁴

Historically defensible for 1987. The SEC staff report characterizes the episode as not merely a one-time revaluation and discusses conditions under which liquidity “disappears” and price limits/coordination issues arise; Brady explicitly treats the episode as market disorder with ad hoc circuit breakers and disconnection. Such conditions are exactly those under which discontinuities may become both more frequent and larger. ³³

Strengths. Purely price-based; relatively interpretable as “nonlinear breakdown” evidence (discrete repricing rather than smooth diffusive adjustment). Some jump methods can output *candidate jump times* at intraday resolution, which helps when you need an explicit timestamp. ²³

Weaknesses. In extremely volatile periods, distinguishing “continuous but huge” volatility from “true jumps” is statistically hard; microstructure noise and recording gaps can masquerade as jumps. High-volatility clustering can also create clusters of detected “jumps” unless you control for intraday periodicity and noise. ³⁴

Can it identify one exact break timestamp? Jump *tests* can identify specific jump minutes, but a single jump minute is not automatically a “market break.” To map jumps into a “break,” you typically need a second layer (e.g., changepoint detection on jump intensity or on a jump-share series) to identify when the *jump regime* begins. ³⁵

Structural break / changepoint detection on returns (and on multivariate feature series)

What it measures. A changepoint algorithm estimates the time(s) at which the statistical properties of a sequence shift—classically the mean/variance in piecewise stationary segments, but more general cost functions can target distributional changes beyond moments. ³⁶

Why it could represent a market break. This is the closest statistical analogue to your conceptual framing: a market break is a **regime shift**, so you estimate the most likely time boundaries between regimes. When primary sources repeatedly emphasize changes in market functioning—interruptions, disconnection, liquidity exhaustion—those are expected to show up as changes in the return-generating process. ³⁰

Historically defensible for 1987. The SEC report's characterization of persistent "aftershocks" and degraded liquidity/continuity suggests that the episode is not only an extreme in magnitude but a shift in market quality. That is compatible with modeling the period as containing distinct statistical regimes. ³⁷

Strengths. Directly yields candidate break timestamps; transparent and widely used; offline algorithms like PELT are computationally efficient and flexible, and reviews emphasize the modular structure (cost function + search method + constraint/penalty) that allows you to tailor detection to "structural disruption" rather than to raw variance. ³⁸

Weaknesses. Plain-vanilla "mean/variance changepoints on returns" can degenerate into "volatility picks," which you explicitly want to avoid. Outliers and data gaps can create false changepoints unless you (i) pre-process carefully and/or (ii) detect breaks on features designed to represent dysfunction (e.g., jump share, staleness rate, serial dependence), rather than on raw squared returns. Penalty choice and short sample length (six trading days) also make "the" break time sensitive without robust checks. ³⁹

Can it identify one exact break timestamp? Yes—this is the main advantage. Both econometric break models and algorithmic changepoint methods output explicit breakpoint indices; for minute data, that corresponds to a minute timestamp (subject to windowing choices if you feed it rolling features). ⁴⁰

Regime shift models in returns (Markov switching / probabilistic regimes)

What it measures. A regime-switching model treats returns as generated by a latent state (e.g., "normal" vs "crisis"), each with its own parameters (mean, variance, possibly AR dynamics), and estimates the probability of being in each state over time. ⁴¹

Why it could represent a market break. A "market break" is naturally described as a transition into a stress regime with different dynamics (higher jumpiness, altered dependence, heavier tails). A regime model gives a probabilistic mapping from time to "state," which is conceptually aligned with "structural disruption." ²⁶

Historically defensible for 1987. Primary accounts describe distinct phases (e.g., severe impairment and near shutdown risk, followed by stabilization). A latent-state framing can be defended as a statistical representation of those phases. ⁴²

Strengths. Provides probabilities (useful when the "break" is not a knife-edge moment); can incorporate multiple features (e.g., AR behavior) within each regime; often robust to noise if specified well. ⁴¹

Weaknesses. With only Oct 16–21 minute data, estimation can be fragile: regime models need enough data per regime to stabilize parameters, and they can collapse back into "variance regimes" if you do not explicitly model discontinuities/serial dependence. They also tend to report *transition probabilities* rather than a single unambiguous timestamp, unless you impose a decision rule (e.g., first time $P(\text{crisis}) > 0.5$). ⁴³

Can it identify one exact break timestamp? Not inherently; it produces a probabilistic transition curve. You can force a timestamp by thresholding the regime probability, but that decision rule is arbitrary and should be reported as such. ⁴³

Staleness / gap / "trading interruption signature" metrics from last price series

What it measures. Rolling fraction of zero returns (unchanged prices), maximum run length of unchanged

price, and explicit missing-minute gap indicators; optionally “locked” behavior (price stuck at a boundary) if such boundaries exist in the contract’s rules for that period.

Why it could represent a market break. Primary accounts treat delayed openings, trading halts, clogged systems, and inability to determine transactable prices as defining features of market dysfunction. Even without volume/quotes, a last-price series can sometimes reflect this via extended staleness or missing intervals corresponding to interruptions or degraded updating. ⁴⁴

Historically defensible for 1987. Stale/delayed prices and impaired information dissemination were central enough that regulators devoted attention to reconstructing trading and to the effects of stale prices on cross-market interpretation. ⁴⁵

Strengths. Extremely simple; robust to the absence of volume/quotes; directly targets “continuity” impairment which is a key non-volatility aspect of market breakdown in the historical narrative. ⁴⁶

Weaknesses. Highly sensitive to how your dataset is constructed (trade price vs indicative price vs last update). “Stale-looking periods” may reflect data vendor sampling gaps rather than true market halts; conversely, active trading can occur without large price moves, producing genuine zero-return runs. This metric is therefore better treated as *supporting evidence* and as a robustness check, not as a stand-alone break definition. ⁴⁷

Can it identify one exact break timestamp? It can flag exact minutes where staleness/gaps begin, but the mapping from that to “market break” is ambiguous unless corroborated by other features (jumps/serial dependence) and/or by a changepoint framework. ⁴⁸

Best defensible proxy for your dataset

Given your constraints (only S&P 500 futures prices; minute sampling; possible missing/stale sections; and the need for a defensible “market break” proxy that is *not* just peak volatility), the most defensible approach is:

Define the “market break” as the first statistically supported changepoint in a multivariate price-only “dysfunction feature” process, where the features are chosen to correspond to the 1987 narrative of impaired functioning—discontinuity, disrupted continuity, and altered microstructure dependence—rather than raw variance alone. ⁴⁹

This choice is historically and conceptually aligned with the primary accounts:

- Brady and GAO emphasize **market disorders, improvisational halts, clogged systems, and disconnection**—all of which plausibly manifest as (i) clusters of discontinuities, (ii) degraded continuity (staleness/gaps), and (iii) unstable serial dependence in the observed price process. ⁵⁰
- The SEC report highlights that volatility is not the whole story and points to **persistent degradation in market quality** after the break (liquidity/continuity/spreads) consistent with a regime shift interpretation, not merely a momentary volatility spike. ⁷
- Changepoint detection is precisely the statistical machinery designed to identify **regime boundaries**, and modern formulations explicitly encourage choosing cost functions/features consistent with the “state” you care about. ⁵¹

From a practical perspective, this is also the most implementable and robust under data issues:

- If missing/stale blocks exist, **feature engineering can be made gap-aware** (compute returns only on consecutive minutes; treat long staleness runs as missing or as their own indicator), and then

changepoints can be detected on the resulting feature series using well-established algorithms like PELT. ⁵²

- This design reduces the risk that your “break” is actually just “the single largest minute return,” because the break is anchored to a **joint shift** in (a) discontinuity/jump share, (b) serial dependence, and (c) continuity/staleness—proxies that map more directly onto the “market functioning impairment” language that dominates the 1987 primary sources. ⁵³

In short: **use volatility-related quantities only insofar as they are components of a broader “structure of returns” shift** (e.g., a rising *jump share* or a changed dependence structure), not as the defining criterion. ⁵⁴

Part B. Action Plan

1) Primary method to test first (recommended):

Build a **multivariate, price-only “dysfunction feature” series** and run **offline changepoint detection (PELT)** to estimate the single most important break timestamp (or the first major break) at 1-minute resolution. Use `ruptures`-style segmentation logic (cost + search + penalty) and *tune the cost/features to detect structural change, not just variance*. ³⁸

2) Backup method to test second:

Run an **intraday jump detection pipeline** and define the market break as the **changepoint in jump intensity / jump share**, not as one isolated jump minute. Concretely: compute a rolling jump-share metric (realized variance vs bipower variation) and detect its first changepoint; optionally corroborate with a jump-time test that outputs candidate jump minutes. ⁵⁵

3) Preprocessing steps you need (gap/staleness-aware):

- Parse timestamps; sort; deduplicate; ensure a strict 1-minute grid per trading session (do *not* forward-fill prices across gaps).
- Create a “is_consecutive_minute” flag and compute log returns **only when timestamps are consecutive and not in a flagged stale block**.
- Identify **stale-looking periods** (e.g., runs of identical prices beyond a chosen threshold) and keep *two versions* of the return series:
 - **Version A:** treat long staleness runs as missing (exclude from return calculations).
 - **Version B:** keep them, but add explicit staleness features (fraction of zero returns, maximum run length).
- Consider resampling to 5-minute returns for jump metrics to reduce microstructure noise sensitivity (then map detected break(s) back to the nearest minute). ⁵⁶

4) Exact outputs to produce (so your result is defensible):

- A single **break timestamp** (minute) from the primary method, plus a short sensitivity band (e.g., how the timestamp changes across reasonable window sizes / penalties).
- Three plots:
 - (i) futures price with vertical line at the break,
 - (ii) each key feature time series (jump share, AR(1)/autocorr, staleness/gap rate) with the same break marker,
 - (iii) a segmentation plot showing the detected regimes.
- A short robustness table: break timestamp under (a) alternative window lengths, (b) “staleness treated as missing” vs “staleness treated as feature,” and (c) primary vs backup method.

5) Main risk/limitation you must explicitly mention in the final explanation:

Your “market break” timestamp is a **futures-only structural break in the observed price process**, and therefore cannot directly validate the historically emphasized **futures-cash linkage failure** (basis dislocation) without cash index data; additionally, missing/stale data can imitate jumps or changepoints, so you must report how you handled these artifacts and show robustness across preprocessing variants. 57

Best Recommendation in One Sentence

Treat the “market break” as the first robust **changepoint in a multivariate, price-only dysfunction feature set** (jump-share + serial dependence + staleness/gap metrics) computed from minute S&P 500 futures prices, rather than as the time of maximum volatility.

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